Lecture 2-1: Point Estimation

Goal: Introduce point estimation and key properties of estimators.

- What is point estimation?
  - Estimands, estimators, estimates
  - Statistical inference
  - The role of sampling
- Properties of estimators
  - Sampling distributions
  - Unbiasedness
  - Efficiency
  - Mean squared error
- Evaluating estimators
  - Analytic derivations
  - Simulation methods
- Asymptotic properties
  - Consistency
  - Convergence in probability
  - Law of large numbers
- Asymptotic normality
  - Central limit theorem
  - Approximate distributions
  - Simulation checks
- Analogy principle
  - Estimating population characteristics with sample counterparts

Worksheet

1. Which of $\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}_3, \hat{\mu}_4$ are unbiased? Show your work.

2. Derive the variances of $\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}_3, \hat{\mu}_4$. Show your work. What do you conclude?

3. Prove: $MSE(\hat{\theta}) = Var(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$.

4. Derive $MSE(\hat{\mu}_1), MSE(\hat{\mu}_2), MSE(\hat{\mu}_3), MSE(\hat{\mu}_4)$